

Dipoles and Its Impacts

Light starts from Dipoles. One can say dipole is one of the most important pieces in nature.

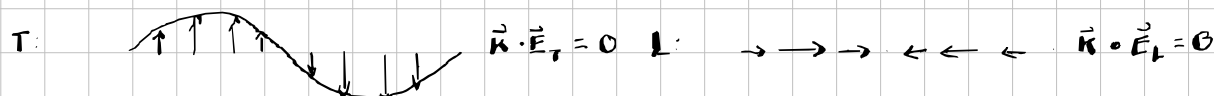
$$\begin{array}{c} q \quad \vec{r} \quad -q \\ \textcircled{+} \quad \longrightarrow \quad \ominus \end{array} \quad \vec{P} = N(-q)\vec{r} \quad N: \text{dipole density.}$$

Dipoles help building electronic transition, lattice vibrations, origin of light, ...

To understand why dipoles are origin of light, look at Macroscopic Maxwell's equations (Gauss unit)

$$\begin{array}{lll} \vec{\nabla} \cdot \vec{E} = -4\pi \vec{\nabla} \cdot \vec{P} + 4\pi \rho_{\text{macro}} & \vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial}{\partial t} \vec{B} & \vec{H} = \vec{B} \\ \vec{\nabla} \cdot \vec{B} = 0 & \vec{\nabla} \times \vec{H} = \frac{1}{c} \frac{\partial}{\partial t} (\vec{E} + 4\pi \vec{P}) + \underbrace{\frac{4\pi}{c} \rho_{\text{macro}}}_0 & \vec{D} = \vec{E} + 4\pi \vec{P} \end{array}$$

Consider simple waves: $\vec{E}(\vec{r}, t) = \vec{E}_T(\vec{r}, t) + \vec{E}_L(\vec{r}, t)$



In Isotropic medium: we have $\vec{E} = \vec{E}_T$, the maxwell's equations give

$$\left\{ \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 \right\} \vec{E}(\vec{r}, t) = -\frac{4\pi}{c^2} \frac{\partial^2}{\partial t^2} \vec{P}(\vec{r}, t) \quad \text{wave equation for transverse wave.}$$

In optical response

$$\vec{P}(\vec{r}, t) = \int_{\text{all}} d^3\vec{r}' \int_{-\infty}^t dt' \tilde{\chi}(\vec{r}, \vec{r}', t, t') \vec{E}(\vec{r}', t') \quad \chi: \text{susceptibility tensor}$$

The response to the fields everywhere and all time contribute to the polarization.

If the material is homogeneous in space and time, that means response is translation invariant

$$\chi(\vec{r} + \vec{a}, \vec{r}' + \vec{a}, t, t') = \chi(\vec{r}, \vec{r}', t, t') \quad (\vec{r} + \vec{a}) - (\vec{r}' + \vec{a}) = \vec{r} - \vec{r}'$$

Since materials look identical everywhere, the physics should not change

$$\begin{array}{ccc} \vec{P}(\vec{r}) & \vec{E}(\vec{r}') & \vec{P}(\vec{r} + \vec{a}) \quad \vec{E}(\vec{r}' + \vec{a}) \\ \uparrow & \leftarrow \uparrow & \uparrow \quad \uparrow \\ & \vec{r} & \vec{r}' \end{array} \Rightarrow \text{the difference } \vec{r} - \vec{r}' \text{ matters}$$

It is also time invariant, since materials at time t are not different to time $t + \Delta t$. Thus,

$$\chi(\vec{r}, \vec{r}', t, t') = \chi(\vec{r} - \vec{r}', t - t')$$

Now, the response simply asks

how far away did the cause occur?

how long ago did the cause occur

$$\vec{P}(\vec{r}, t) = \int d^3\vec{r}' \int_{-\infty}^t dt' \tilde{\chi}(\vec{r} - \vec{r}', t - t') \vec{E}(\vec{r}', t') \Leftrightarrow \vec{P}(\vec{r}, \omega) = \chi(\vec{r}, \omega) \vec{E}(\vec{r}, \omega)$$

This is now a convolution in space and time.

If the material is isotropic, no direction is preferred.

$$\tilde{\chi} = \chi \mathbf{I} \quad P_x \Leftrightarrow E_x \quad P_y \Leftrightarrow E_y \quad P_z \Leftrightarrow E_z$$

Dipoles are created by E field and in turn creates E field. But the response χ needs some models to predict

Relations Between Lights and Materials

From the wave equations

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2\right) \vec{E}(\vec{r}, t) = -\frac{4\pi}{c^2} \frac{\partial^2}{\partial t^2} \vec{P}(\vec{r}, t)$$

Since $\mathcal{F}\left[\frac{\partial^2}{\partial t^2} \vec{E}\right] = -\omega^2 \vec{E}$ and $\mathcal{F}[\nabla^2 \vec{E}] = -k^2 \vec{E}$ and $\mathcal{F}[\vec{P}(\vec{r}, t)] = \chi(\vec{k}, \omega) \vec{E}(\vec{k}, \omega)$

$$\left\{\frac{\omega^2}{c^2} (1 + 4\pi \chi(\vec{k}, \omega)) - k^2\right\} \vec{E}(\vec{k}, \omega) = 0$$

From a propagating E field where material creates a polarization source for a new additional generated field, we arrive at the FT. It analysis how the material reacts to different light frequencies

$$\boxed{\frac{\omega^2}{c^2} \epsilon(\vec{k}, \omega) = k^2} \quad \text{where } \epsilon = 1 + 4\pi \chi(\vec{k}, \omega) = n^2 \quad \text{Dispersion}$$

Now, assume a mono chromatic comes in the medium

$$\vec{E}(z, t) = \frac{1}{2} \vec{E}_0 e^{-i(\omega t - kz)} + c.c. \quad \omega \text{ is real and } k \text{ complex.} \quad \text{Im } k > 0 \text{ decay} \quad \text{Im } k < 0 \text{ grow}$$

Insert in wave equation gives

$$\frac{\omega^2}{c^2} \epsilon = k^2 \rightarrow \text{exactly as above and this form } e^{\pm i} \text{ is a solution}$$

We learn that k is complex when ϵ is complex

$$k = k' + i k'' \quad \epsilon = \epsilon' + i \epsilon''$$

Expanding gives

$$\frac{\omega^2}{c^2} \epsilon' = k'^2 - k''^2 \quad \frac{\omega^2}{c^2} \epsilon'' = 2k' k''$$

Define

$$k_0 = \frac{\omega}{c} \quad k' = n k_0 \quad \alpha = 2k'' \quad \text{By observing the } E \text{ form, } k' \text{ plays an oscillatory role while } k'' \text{ represents the absorption}$$

$$\vec{E}(z, t) = \frac{1}{2} \vec{E}_0 e^{-i(\omega t - n k_0 z)} e^{-\alpha z/2} + c.c.$$

Alternative definition

$$\tilde{n} = n + i k \\ n = \sqrt{\epsilon' + i \epsilon''} \\ k = \frac{\omega}{c} \tilde{n}$$

Solving:

$$n = \sqrt{\frac{1}{2} [\epsilon' + \sqrt{\epsilon'^2 + \epsilon''^2}]} \quad \alpha = \frac{\omega}{c n} \epsilon''$$

By taking intensity, we arrive right at the Beer's law

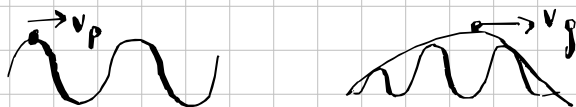
$$\text{Intensity} \sim \left| \frac{1}{2} \vec{E}_0 e^{-i(\omega t - kz)} \right|^2 \\ \sim e^{-\alpha z}$$

Consider a pulse, it has many nearby frequencies $\omega_0, \omega_0 \pm \delta\omega, \dots$ or $k_0, k_0 \pm \delta k$. For example

$$E_1 = \cos(k_1 z - \omega_1 t) \quad \text{and} \quad E_2 = \cos(k_2 z - \omega_2 t)$$

$$E = 2 \cos(\Delta k z - \Delta \omega t) \cos(\bar{k} z - \bar{\omega} t) \quad \bar{k} = \frac{k_1 + k_2}{2}$$

$$v_p = \frac{\omega}{k} \quad v_g = \frac{\Delta \omega / \Delta k}{\Delta \omega / \Delta k} = \frac{\partial \omega}{\partial k}$$



In vacuum, $\omega = ck$ $v_g = c$, phase velocity is the same as group velocity (non dispersive)

When material is dispersive, $k(\omega) = \frac{\omega n(\omega)}{c} \Rightarrow v_g = \frac{c}{n + \omega \frac{dn}{d\omega}}$

The group velocity is different than phase velocity.